STEP II OVER AN ARBITRARY COEFFICIENT RING

Abstract. Let A be any commutative ring, $F, G, U, V \in A[X, Y]$ with $FU = GV$, and J the Jacobian of F, G. We show $J^e U \in (G)$ for some e, without the annihilator stabilization hypothesis of `jacobian-regularity.pdf` and without passing to a noetherian coefficient ring by a classical argument. Noetherianity is used once, along one sequence, in the constructive form of Coquand–Persson, inside a subring of $B = (A[X, Y]/(G))[1/J]$ that is given by codes.

Throughout, $B = (A[X, Y]/(G))[1/J]$, and x, y, f, u are the images of X, Y, F, U in B. As in `jacobian-regularity.pdf`, it suffices to prove $u = 0$ in B: the kernels of localization and of the quotient by G then give $J^e U = GW$ in $A[X, Y]$.

1. The kill argument

For each n, finite Taylor lifting gives $x_n(T), y_n(T) \in B[T]$ with $x_n(0) = x$, $y_n(0) = y$, $F(x_n, y_n) \equiv f - T$ and $G(x_n, y_n) \equiv 0$ modulo T^{n+1}. Substituting in $FU = GV$ gives $(f - T)\, p_n(T) \equiv 0 \pmod{T^{n+1}}$ with $p_n = U(x_n, y_n)$. Comparing coefficients, with w_n the coefficient of T^n in p_n,

$$u = p_n(0) = f^n w_n, \qquad f^{n+1} w_n = 0.$$

Lemma 1.1 (Kill)**.** *Let $u, f, w_0, w_1, \ldots$ be elements of a ring with $u = f^n w_n$ and $f^{n+1} w_n = 0$ for all n. If some w_k is a linear combination of $w_0, \ldots, w_{k-1}$, then $u = 0$.*

Proof. For $j < k$, $f^k w_j = f^{k-1-j} f^{j+1} w_j = 0$, so f^k kills w_k, and $u = f^k w_k = 0$. For $k = 0$ the empty combination gives $w_0 = 0 = u$. □

So it suffices that the sequence (w_n) has a *good* initial segment: one whose last element is a combination of the earlier ones. In a noetherian ring in the sense of Coquand–Persson ("good bars the empty list") every sequence has one, provided the elements of the sequence can be followed along the tree of the bar; this needs a code for each w_n, but only merely, since the conclusion $u = 0$ is a proposition.

2. The coded subring

Let $a_1, \ldots, a_m$ be the coefficients of F, G, U, V, and let $A_0 \subseteq A$ be the subring they generate. Its image in B, together with x, y and $1/J$, generates a subring of B that is a quotient of $\mathbb{Z}[t_1, \ldots, t_{m+3}]$; we present it by codes.

Definition 2.1. Let K be the type of codes of $\mathbb{Z}[t_1, \ldots, t_{m+3}]$ (iterated polynomial codes over the codes of $\mathbb{Z}$), and $\varphi : K \to B$ the evaluation at $h(a_1), \ldots, h(a_m), x, y, 1/J$. It preserves the ring operations; no law is asked of K. The *coded subring* is

$$B' = \sum_{b:B} \left\| \textstyle\sum_{c:K} \varphi(c) = b \right\|.$$

The predicate is a proposition, so B' is a set-level Σ and the projection $\iota : B' \to B$ is injective: elements of B' are equal when their images are. Coded elements are closed under the ring operations (apply the operation to codes, under the truncation), so B' is a commutative

Date: October 26, 2026.

ring, every law holding because it holds in B, and ι is a ring map. The elements $h(a_i)$, x, y, $1/J$ lie in B', with the codes t_i.

Lemma 2.2. *B' is* coded noetherian*: the presentation $\varphi' : K \to B'$, $\varphi'(c) = (\varphi(c), \|(c, \mathrm{refl})\|)$, is noetherian, and every element of B' merely has a code.*

Proof. The second part is the definition of B'. For the first, φ is noetherian by the Hilbert basis theorem iterated $m + 3$ times over the image of $\mathbb{Z}$ (`sag/noetherian`). A combination in B' with coefficients in K is one in B with the same coefficients, and conversely, since ι is injective and $\iota \circ \varphi' = \varphi$. So the good lists for φ' and φ are the same lists of codes, and the tree of the bar for φ is one for φ'. □

3. Running the lifting in B'

Write F', G', U', V' for F, G, U, V with coefficients mapped to B' (through $A_0 \to B'$). In B' we have $G'(x, y) = 0$ and $J'(x, y)$ invertible with inverse $1/J$: both equations hold after ι, hence in B'. So Taylor lifting can be run in B', giving $x'_n, y'_n \in B'[T]$, $p'_n = U'(x'_n, y'_n)$ and

$$u' = {f'}^n w'_n, \qquad {f'}^{n+1} w'_n = 0 \qquad \text{in } B',$$

with $\iota(u') = u$. The argument of the previous section uses the relation $FU = GV$ only through its consequence

$$F'(x'_n, y'_n)\, U'(x'_n, y'_n) = G'(x'_n, y'_n)\, V'(x'_n, y'_n) \qquad \text{in } B'[T].$$

This is not obtained from a relation in $A_0[X, Y]$. Apply ι to coefficients: $\iota[T] : B'[T] \to B[T]$ is a ring map, and carries $H'(x'_n, y'_n)$ to $H(\iota x'_n, \iota y'_n)$ for every polynomial H over A. In $B[T]$ the identity holds, as the image of $FU = GV$ under the A-algebra map $A[X, Y] \to B[T]$, $X \mapsto \iota x'_n$, $Y \mapsto \iota y'_n$. Since ι is injective, the two sides agree coefficientwise in B'.

4. Conclusion

Each w'_n is an element of B', so it merely has a code: this is the point of running the lifting in B' rather than in B, where the coefficients of the lift are built from x, y, $1/J$ and the a_i by the ring operations but this is not recorded. By Lemma 2.2 the walk along the tree of the bar, following merely given codes of $w'_0, w'_1, \ldots$, merely finds a good initial segment, and Lemma 1.1 gives $u' = 0$, a proposition, so the truncation is eliminated without choice. Then $u = \iota(u') = 0$ in B.

Theorem 4.1. *For every commutative ring A and $F, G, U, V \in A[X, Y]$ with $FU = GV$, merely $J^e U = GW$ for some $e \geq 0$ and $W \in A[X, Y]$.*

No excluded middle, choice, decidable equality, or noetherian hypothesis on A is used: the only noetherian rings are the finitely generated ones over $\mathbb{Z}$, presented by codes.

Formalisation

In `unit/sag/jacobian`: `CodedSubring` (the subring B' of a presentation, its ring structure, Lemma 2.2, and Lemma 1.1 along a merely coded sequence, from `Sequence.Kill.Walk`); `RawMap` (a ring map on coefficient lists); `RegularAt.Curve` (the order-n calculation from the curve identity alone); `FiniteCoefficients` (the generators, F', G', U', V', the transfer of $G(x, y) = 0$, of $1/J$ and of the curve identity along ι, and the theorem). In the formal text the coefficient map is the identity of B': only the images $h(a_i)$ of the generators of A_0 are needed, so A_0 is not built as a ring of its own.